2 **(a)** State Newton's law of gravitation

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..... [1]

(b) A satellite with a mass 6200 kg is at point X, a distance of 1.50×10^9 m from the centre of the Earth, as shown in Fig. 2.1.

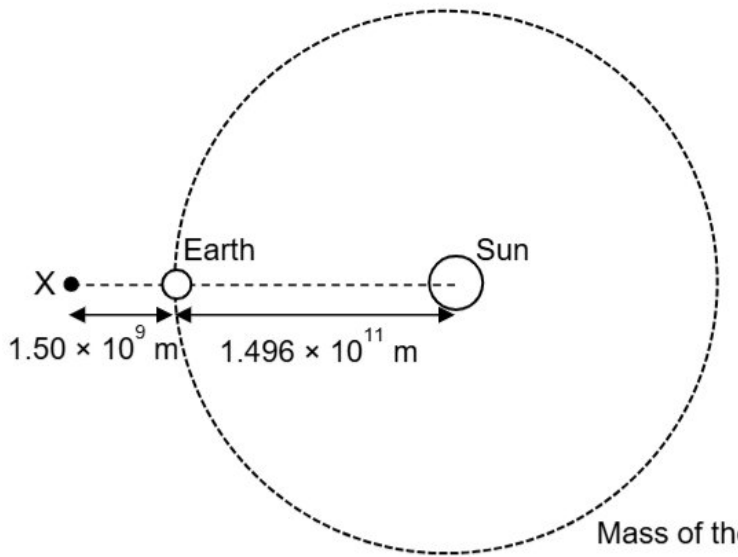


Fig. 2.1
(not to scale)

Mass of the Sun = 1.99×10^{30} kg

Mass of the Earth = 5.98×10^{24} kg

Earth-Sun distance = 1.496×10^{11} m

(i) Determine the magnitude of the resultant force acting on the satellite due to the Sun and the Earth.

resultant force = ...N [3]

(ii) The satellite is in a circular orbit around *the Sun*. Calculate the angular velocity of the satellite.

angular velocity = ... rad s^{-1} [2]

(iii) The angular velocity of the Earth in its orbit around the Sun is known to be 1.99×10^{-7} rad s^{-1}. Using your answer to **(b)(ii),** describe the motion of the satellite relative to the Earth.

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..... [1]

(c) A satellite of mass m orbits the Earth in circular path as shown in Fig. 2.2. The satellite has a speed v and the radius of its orbit is r.

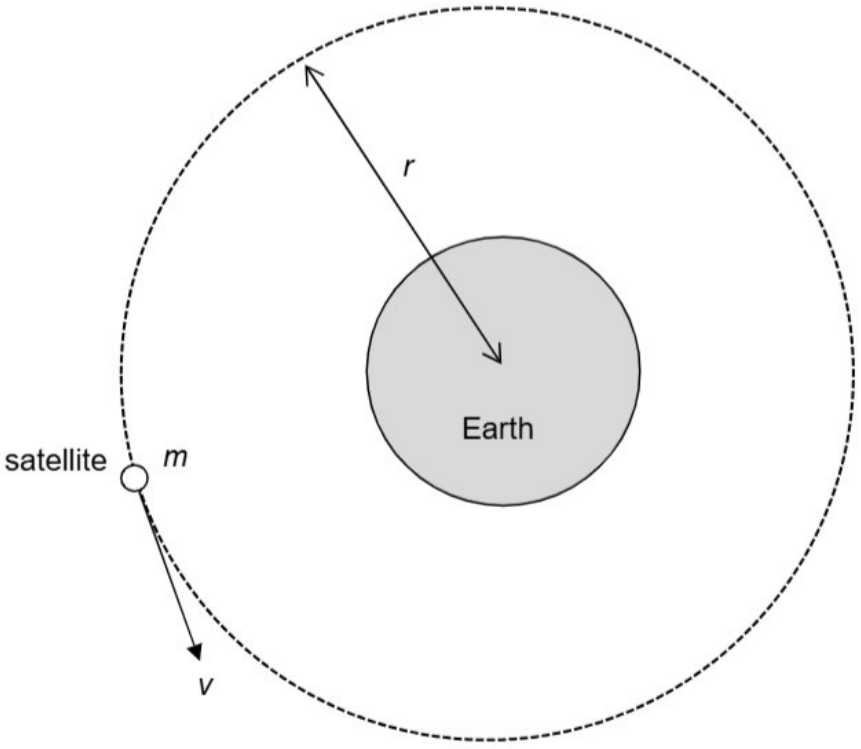


Fig. 2.2

The speed v is related to the radius r by the relation $v = kr^n$.

Determine the expression for k and n in terms of G, M and m, where M is the mass of the Earth and G is the gravitational constant.

k = ...

n = ... [3]

(d) A satellite moves in a circular orbit around the Earth. Its engines are fired briefly in a direction that is opposite to its motion, causing its speed to decrease.

Explain why the satellite subsequently moves towards the Earth instead of continuing in the same circular orbit at a lower speed.

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..... [2]